Question Slides

LaTeXQuest

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Question 1

Example 1. Find the values of

(i) ${}^5P_5$

(ii) ${}^5P_3$

(iii) ${}^{60}P_{48}$

(iv) ${}^{60}P_2$

[1.0 mark]

Solution 1

Using the permutation formula ${}^nP_r = \dfrac{n!}{(n-r)!}$:

(i) ${}^5P_5 = \dfrac{5!}{(5-5)!} = \dfrac{5!}{0!} = 5! = 120.$

(ii) ${}^5P_3 = \dfrac{5!}{(5-3)!} = \dfrac{5!}{2!} = 5 \times 4 \times 3 = 60.$

(iii) ${}^{60}P_{48} = \dfrac{60!}{(60-48)!} = \dfrac{60!}{12!}.$

(iv) ${}^{60}P_2 = \dfrac{60!}{(60-2)!} = 60 \times 59 = 3540.$

Question 2

Example 2. Find n if

(i) ${}^{n}P_4 : {}^{n}P_5 = 1 : 2$

(ii) ${}^{n}P_4 : {}^{n-1}P_3 = 9 : 1$

(iii) $P(n, 4) = 20\,P(n, 2)$

[1.0 mark]

Solution 2

(i) $\dfrac{{}^{n}P_4}{{}^{n}P_5} = \dfrac{\frac{n!}{(n-4)!}}{\frac{n!}{(n-5)!}} = \dfrac{n-5!}{(n-4)!} = \dfrac{1}{n-4} = \dfrac{1}{2} \implies n-4 = 2 \implies n = 6.$

(ii) $\dfrac{{}^{n}P_4}{{}^{n-1}P_3} = \dfrac{\frac{n!}{(n-4)!}}{\frac{(n-1)!}{(n-4)!}} = \dfrac{n!}{(n-1)!} = n = \dfrac{9}{1} \implies n = 9.$

(iii) $P(n,4) = n(n-1)(n-2)(n-3)$ and $P(n,2) = n(n-1)$. Since $n \geq 4$, dividing both sides by $n(n-1)$ gives: $(n-2)(n-3) = 20 \implies n^2 - 5n + 6 = 20 \implies n^2 - 5n - 14 = 0 \implies (n-7)(n+2) = 0$. Since n must be a positive integer, $n = 7$.

Question 3

Example 4. Find r if:

(i) $^{9}P_{5} + 5\,^{9}P_{4} = {}^{10}P_{r}$

(ii) $^{22}P_{r+1} : {}^{20}P_{r+2} = 11 : 52$

[1.0 mark]

Solution 3

1. By the identity ${}^nP_r + r \cdot {}^nP_{r-1} = {}^{n+1}P_r$, with $n = 9$ and $r = 5$:

$$ {}^9P_5 + 5\,{}^9P_4 = {}^{10}P_5. $$

Therefore, ${}^{10}P_r = {}^{10}P_5 \implies r = 5$.

2. We have:

$$ \frac{\frac{22!}{(21-r)!}}{\frac{20!}{(18-r)!}} = \frac{22 \times 21}{(21-r)(20-r)(19-r)} = \frac{11}{52}. $$

Dividing by 11:

$$ \frac{2 \times 21}{(21-r)(20-r)(19-r)} = \frac{1}{52} \implies (21-r)(20-r)(19-r) = 42 \times 52 = 2184. $$

Factoring 2184 into 3 consecutive integers: $2184 = 14 \times 13 \times 12$. Thus, $21 - r = 14 \implies r = 7$.

Question 4

Example 5. Find n if:

(i) ${}^{n}P_{5} = 42\,{}^{n}P_{3}$

(ii) ${}^{2n+1}P_{n-1} : {}^{2n-1}P_{n} = 3 : 5$

[1.0 mark]

Solution 4

1. For $n \geq 5$:

$$\frac{n!}{(n-5)!} = 42\,\frac{n!}{(n-3)!} \implies (n-3)(n-4) = 42.$$

Since $42 = 7 \times 6$, we have $n - 3 = 7 \implies n = 10$.

2. Expanding the ratio:

$$\frac{\frac{(2n+1)!}{(n+2)!}}{\frac{(2n-1)!}{(n-1)!}} = \frac{(2n+1)(2n)}{(n+2)(n+1)n} = \frac{2(2n+1)}{(n+2)(n+1)} = \frac{3}{5}.$$

Cross-multiplying:

$$10(2n+1) = 3(n^2+3n+2) \implies 20n+10 = 3n^2+9n+6 \implies 3n^2-11n-4 = 0.$$

Factoring: $(3n+1)(n-4) = 0$. Since $n \in \mathbb{N}$, $n = 4$.

Question 5

Example 6. Find the value of r if

(i) $5\,{}^4P_r = 6\,{}^5P_{r-1}$ (NCERT)

(ii) $P(10, r+1) : P(11, r) = 30 : 11$

[1.0 mark]

Solution 5

(i) $5 \cdot \dfrac{4!}{(4-r)!} = 6 \cdot \dfrac{5!}{(6-r)!} \implies \dfrac{5 \times 24}{(4-r)!} = \dfrac{6 \times 120}{(6-r)(5-r)(4-r)!}$.

Dividing both sides by $\dfrac{120}{(4-r)!}$ gives:

$1 = \dfrac{6}{(6-r)(5-r)} \implies (6-r)(5-r) = 6 = 3 \times 2.$

Thus, $6 - r = 3 \implies r = 3$ (or $6 - r = -2 \implies r = 8$, which is rejected since $r \leq 4$). Hence, $r = 3$.

(ii) $\dfrac{\frac{10!}{(9-r)!}}{\frac{11!}{(11-r)!}} = \dfrac{(11-r)(10-r)}{11} = \dfrac{30}{11} \implies (11-r)(10-r) = 30 = 6 \times 5.$

Hence, $11 - r = 6 \implies r = 5$.

Question 6

Example 7. Prove that

(i) $P(2n, n) = 2^n[1 \cdot 3 \cdot 5 \cdots (2n-1)]$

(ii) $P(n, r) = n \cdot P(n-1, r-1)$

(iii) ${}^nP_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

[1.0 mark]

Solution 6

(i) $P(2n, n) = \dfrac{(2n)!}{n!} = \dfrac{[1 \cdot 3 \cdot 5 \cdots (2n-1)] \times [2 \cdot 4 \cdot 6 \cdots 2n]}{n!} = \dfrac{[1 \cdot 3 \cdots (2n-1)] \times 2^n (n!)}{n!} = 2^n [1 \cdot 3 \cdot 5 \cdots (2n-1)].$

(ii) $n \cdot P(n-1, r-1) = n \cdot \dfrac{(n-1)!}{((n-1)-(r-1))!} = \dfrac{n(n-1)!}{(n-r)!} = \dfrac{n!}{(n-r)!} = P(n, r).$

(iii) ${}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1} = \dfrac{(n-1)!}{(n-1-r)!} + \dfrac{r(n-1)!}{(n-r)!} = \dfrac{(n-1)!}{(n-r)!}[(n-r)+r] = \dfrac{n(n-1)!}{(n-r)!} = \dfrac{n!}{(n-r)!} = {}^nP_r.$

Question 7

Example 8. How many 4-letter words, with or without meaning, can be formed out of the letters of the word WONDERFUL, if repetition of letters is not allowed?

[1.0 mark]

Solution 7

The word WONDERFUL contains 9 distinct letters (W, O, N, D, E, R, F, U, L). The number of 4-letter words is:

$$^9P_4 = 9 \times 8 \times 7 \times 6 = 3024.$$

Example 9. How many words, with or without meaning, can be formed by using all the letters of the word EQUATION, using each letter exactly once?

[1.0 mark]

Solution 8

The word EQUATION contains 8 distinct letters (E, Q, U, A, T, I, O, N). The number of permutations using all 8 letters is:

$$8! = 40320.$$

Example 10. A customer forgets a four-digit code for an Automatic Teller Machine (ATM) in a bank. However, he remembers that this code consists of 3, 4, 7 and 9. Find the largest possible number of trials necessary to obtain the correct code.

[1.0 mark]

Solution 9

The number of possible four-digit codes formed from the 4 distinct digits 3, 4, 7, 9 is:

$$4! = 24.$$

Thus, the maximum number of trials required is 24.

Example 11. How many 4-digit numbers can be formed by using the digits 1 to 9 if repetition of digits is not allowed?

[1.0 mark]

Solution 10

There are 9 available digits $(1, 2, \ldots, 9)$. Choosing and arranging 4 digits gives:

$$^{9}P_{4} = 9 \times 8 \times 7 \times 6 = 3024.$$

Example 12. From a sports committee of 12 persons, in how many ways can you choose a president and a vice president assuming that one person cannot hold more than one office?

[1.0 mark]

Solution 11

The president can be chosen in 12 ways, and the vice president from the remaining 11 persons in 11 ways:

$$^{12}P_2 = 12 \times 11 = 132 \text{ ways.}$$

Example 13. Ten students participate in a debate. In how many ways can the first three prizes be won?

[1.0 mark]

Solution 12

Selecting and ordering 3 winners out of 10 students:

$$^{10}P_3 = 10 \times 9 \times 8 = 720 \text{ ways.}$$

Example 14. In how many ways can three sports prizes be given to 20 boys when a boy may receive any number of prizes?

[1.0 mark]

Solution 13

Since each prize can be awarded to any of the 20 boys independently:

$$20 \times 20 \times 20 = 20^3 = 8000 \text{ ways.}$$

Example 15. Find the total number of ways in which n distinct balls can be put into two different boxes so that no box remains empty.

[1.0 mark]

Solution 14

Each of the n distinct balls can be placed into either of the 2 boxes, giving 2^n total assignments. The 2 excluded cases are when all balls are in Box 1 or all are in Box 2 (leaving one box empty). Thus, the required number of ways is:

$$2^n - 2.$$

Example 16. A question paper contains 10 true-false questions. In how many ways can a student attempt one or more questions?

[1.0 mark]

Solution 15

For each question, there are 3 possibilities: True, False, or Unattempted. Total possible ways to answer the paper $= 3^{10}$. Subtracting the 1 way where no question is attempted gives:

$$3^{10} - 1 = 59049 - 1 = 59048 \text{ ways.}$$

Example 17. Find the number of signals which can be given with 5 flags of different colours hoisted one above the other, by using any number of flags.

[1.0 mark]

Solution 16

Signals can be made using 1, 2, 3, 4, or 5 flags:

$$\begin{aligned}\text{Total} &= {}^5P_1 + {}^5P_2 + {}^5P_3 + {}^5P_4 + {}^5P_5 \\ &= 5 + 20 + 60 + 120 + 120 = 325 \text{ signals.}\end{aligned}$$

Question 17

Example 18. How many words can be made from the letters in the word MONDAY, assuming that no letter is repeated, if

(i) 4 letters are used at a time?

(ii) all letters are used at a time?

(iii) all letters are used but the first letter is a vowel?

[1.0 mark]

Solution 17

The word MONDAY contains 6 distinct letters: 2 vowels (O, A) and 4 consonants (M, N, D, Y).

(i) ${}^6P_4 = 6 \times 5 \times 4 \times 3 = 360$.

(ii) ${}^6P_6 = 6! = 720$.

(iii) The first place can be filled by any of the 2 vowels in 2 ways. The remaining 5 places can be filled by the remaining 5 letters in $5! = 120$ ways. Total $= 2 \times 120 = 240$.

Question 18

Example 19. Find the number of ways of arranging the letters of the word GARDEN such that the vowels remain in alphabetical order.

[1.0 mark]

Solution 18

The word GARDEN has 6 distinct letters with vowels A and E. Total permutations of 6 letters is $6! = 720$. In exactly half of these arrangements, A precedes E:

$$\frac{6!}{2!} = \frac{720}{2} = 360 \text{ ways.}$$

Question 19

Example 20. How many different words can be formed with the letters of the word ORDINATE so that vowels occupy odd places?

[1.0 mark]

Solution 19

ORDINATE has 8 letters: 4 vowels (O, I, A, E) and 4 consonants (R, D, N, T). There are 4 odd positions (1st, 3rd, 5th, 7th). The 4 vowels can occupy these 4 positions in 4! ways. The 4 consonants occupy the remaining 4 even positions in 4! ways:

$$4! \times 4! = 24 \times 24 = 576.$$

Question 20

Example 22. How many words can be formed out of the letters of the word ORIENTAL so that A and E occupy odd places?

[1.0 mark]

Solution 20

ORIENTAL has 8 distinct letters. The odd positions are 1, 3, 5, 7 (4 available positions). The 2 letters A and E can be placed in any of the 4 odd places in ${}^4P_2 = 12$ ways. The remaining 6 letters can be arranged in the remaining 6 positions in $6! = 720$ ways:

$$ {}^4P_2 \times 6! = 12 \times 720 = 8640. $$

Example 23. How many words with or without meaning can be formed with the letters of the word EQUATION so that the vowels and consonants occur together? (NCERT)

[1.0 mark]

Solution 21

In EQUATION, there are 5 vowels (E, U, A, I, O) and 3 consonants (Q, T, N). Group the vowels as one block and consonants as another block. The 2 blocks can be arranged in $2! = 2$ ways. The 5 vowels inside their block can be arranged in $5! = 120$ ways, and the 3 consonants in $3! = 6$ ways:

$$2! \times 5! \times 3! = 2 \times 120 \times 6 = 1440.$$

Question 22

Example 24. In how many ways 3 mathematics books, 4 history books, 3 chemistry books and 2 biology books can be arranged on a shelf so that all the books on the same subject are together? (Exemplar)

[1.0 mark]

Solution 22

Treat each subject group as a single unit: 4 distinct units (Math, History, Chemistry, Biology). The 4 units can be arranged in 4! ways. Books within each subject can be arranged among themselves:

$$4! \times 3! \times 4! \times 3! \times 2! = 24 \times 6 \times 24 \times 6 \times 2 = 41472 \text{ ways.}$$

Question 23

Example 25. Three married couples are to be seated in a row having six seats in a cinema hall. If spouses are to be seated next to each other, in how many ways can they be seated? Find also the number of ways of their seating if all the ladies sit together and all gents sit together. (Exemplar)

[1.0 mark]

Solution 23

(a) *Spouses seated next to each other:* Treat each couple as a single unit. There are 3 units, arranged in $3! = 6$ ways. Each couple can interchange their seats in $2! = 2$ ways:

$$3! \times (2!)^3 = 6 \times 8 = 48 \text{ ways.}$$

(b) *All ladies together and all gents together:* There are 2 units (the block of 3 ladies and the block of 3 gents), arranged in $2! = 2$ ways. The ladies can be arranged in $3! = 6$ ways and gents in $3! = 6$ ways:

$$2! \times 3! \times 3! = 2 \times 6 \times 6 = 72 \text{ ways.}$$

Example 26. 7 candidates are to be examined—2 in Mathematics and the remaining in different subjects. In how many ways can they be seated in a row so that the two examinees in Mathematics may not sit together?

[1.0 mark]

Solution 24

First arrange the 5 other candidates in a row: $5! = 120$ ways. This creates 6 available gaps (including ends): $_C_1_C_2_C_3_C_4_C_5_$. The 2 Mathematics candidates can be placed in any 2 of these 6 gaps in ${}^6P_2 = 30$ ways:

$$5! \times {}^6P_2 = 120 \times 30 = 3600 \text{ ways.}$$

Question 25

Example 27. Ten different letters of an alphabet are given. Find the number of five-letter words that can be formed from the given letters which have at least one letter repeated. (Exemplar)

[1.0 mark]

Solution 25

Total number of 5-letter words with repetition allowed $= 10^5 = 100000$. Total number of 5-letter words with no repetition $= {}^{10}P_5 = 10 \times 9 \times 8 \times 7 \times 6 = 30240$. Number of words with at least one letter repeated:

$$10^5 - {}^{10}P_5 = 100000 - 30240 = 69760.$$

Example 28. Find the number of different words that can be formed from the letters of the word TRIANGLE so that no two vowels are together. (Exemplar)

[1.0 mark]

Solution 26

TRIANGLE has 8 distinct letters: 5 consonants (T, R, N, G, L) and 3 vowels (I, A, E). Arrange the 5 consonants first in $5! = 120$ ways. This creates 6 gaps: ${}^6P_3$ ways to arrange the 3 vowels in these gaps:

$$5! \times {}^6P_3 = 120 \times (6 \times 5 \times 4) = 120 \times 120 = 14400.$$

Example 29. Five boys and 5 girls are to be seated on a bench with the boys and girls alternately. Find the number of ways of their seating.

[1.0 mark]

Solution 27

Two alternate patterns are possible:

1. B G B G B G B G B G
2. G B G B G B G B G B

For each pattern, boys can be seated in $5! = 120$ ways and girls in $5! = 120$ ways.

$$\text{Total} = 2 \times (5! \times 5!) = 2 \times 120 \times 120 = 28800 \text{ ways.}$$

Question 28

Example 30. How many different (eight-letter) words can be formed out of the letters of the word DAUGHTER so that

(i) the word starts with D and ends with R

(ii) position of letter H remains unchanged

(iii) relative position of vowels and consonants remains unaltered

(iv) no two vowels are together

[1.0 mark]

Solution 28

The word DAUGHTER has 8 letters: 3 vowels (A, U, E) and 5 consonants (D, G, H, T, R).

(i) Fixing D at position 1 and R at position 8 leaves 6 letters for 6 positions: $6! = 720$.

(ii) Fixing H in its original 5th position leaves 7 letters for 7 positions: $7! = 5040$.

(iii) Vowels must stay in the original vowel positions (2nd, 3rd, 7th) and consonants in original consonant positions (1st, 4th, 5th, 6th, 8th): $3! \times 5! = 6 \times 120 = 720$.

(iv) Place the 5 consonants first ($5! = 120$). In the 6 available gaps, place the 3 vowels (${}^6P_3 = 120$): $120 \times 120 = 14400$.

Question 29

Example 31. Delhi's car licence plate consists of 8 figures — 'D' followed by an alphabet followed by a numeral 1 to 4 followed by letter A, B, C or D followed by a four-digit non-zero number. How many licence plates can be made without duplicating?

[1.0 mark]

Solution 29

Let the format be $D - L_1 - N_1 - L_2 - \text{Number(4-digit)}$.

- Fixed letter: 'D' $\implies$ 1 way.
- Alphabet L_1: 26 possibilities.
- Numeral $N_1 \in \{1, 2, 3, 4\}$: 4 possibilities.
- Letter $L_2 \in \{\text{A, B, C, D}\}$: 4 possibilities.
- 4-digit non-zero number from 0001 to 9999: $10^4 - 1 = 9999$ possibilities.

Total number of plates:

$$1 \times 26 \times 4 \times 4 \times 9999 = 416 \times 9999 = 4159584.$$

Example 32. How many 6-digit numbers can be formed from the digits 0, 1, 3, 5, 7, 9 which are divisible by 10 and no digit is repeated? (NCERT)

[1.0 mark]

Solution 30

For a number to be divisible by 10, the units digit must be 0. The remaining 5 digits (1, 3, 5, 7, 9) can fill the first 5 places in:

$$5! = 120 \text{ numbers.}$$

Question 31

Example 33. Find the number of 4-digit numbers that can be formed with the digits 1, 2, 3, 4, 5 if no digit is used more than once in a number. How many of these numbers will be even?

[1.0 mark]

Solution 31

(a) Total 4-digit numbers using 1, 2, 3, 4, 5:

$$^5P_4 = 5 \times 4 \times 3 \times 2 = 120.$$

(b) For even numbers, the unit place must be filled by 2 or 4 (2 choices). The remaining 3 places are filled from the remaining 4 digits in $^4P_3 = 24$ ways:

$$2 \times 24 = 48 \text{ even numbers.}$$

Example 34. How many numbers greater than 50000 can be formed by using the digits 0, 2, 3, 5 and 6, each digit is used only once in each number?

[1.0 mark]

Solution 32

A 5-digit number greater than 50000 must start with 5 or 6 (2 choices for the ten-thousands place). The remaining 4 places can be arranged in $4! = 24$ ways:

$$2 \times 4! = 2 \times 24 = 48 \text{ numbers.}$$

Question 33

Example 35. Find the number of integers greater than 7000 that can be formed with the digits 3, 5, 7, 8 and 9 where no digits are repeated. (Exemplar)

[1.0 mark]

Solution 33

The numbers can be 4-digit or 5-digit:

- *4-digit numbers* > 7000*:* The thousands place can be filled by 7, 8, or 9 (3 choices). The remaining 3 places are filled from the 4 remaining digits in ${}^4P_3 = 24$ ways. Total 4-digit $= 3 \times 24 = 72$.
- *5-digit numbers:* Any 5-digit number formed from these digits is > 7000. Total 5-digit $= 5! = 120$.

Total integers $= 72 + 120 = 192$.

Example 36. How many 5-digit numbers divisible by 3 can be formed by using the digits 0, 1, 2, 3, 4 and 5 without repetition of digits? (Exemplar)

[1.0 mark]

Solution 34

The sum of all six digits is $0+1+2+3+4+5=15$. To choose 5 digits whose sum is divisible by 3, the excluded digit must be a multiple of 3 (either 0 or 3):

- *Case 1: Exclude 0.* Digits are $\{1,2,3,4,5\}$. Number of 5-digit numbers $= 5! = 120$.
- *Case 2: Exclude 3.* Digits are $\{0,1,2,4,5\}$. The first digit cannot be 0 (4 choices), and the remaining 4 places can be arranged in $4! = 24$ ways. Number of 5-digit numbers $= 4 \times 4! = 96$.

Total $= 120 + 96 = 216$.

Example 38. Find the number of 4-digit numbers greater than 4000 whose sum of digits is odd.

[1.0 mark]

Solution 35

The 4-digit numbers greater than 4000 range from 4001 to 9999. Total numbers in this range $= 9999 - 4000 = 5999$. Consider the numbers from 4000 to 9999 (a total of 6000 consecutive integers). For every block of 10 consecutive numbers ending in $0, 1, 2, \ldots, 9$, exactly half (5) have an odd digit sum and half (5) have an even digit sum. Thus, in the 6000 integers from 4000 to 9999:

$$\text{Numbers with odd digit sum} = \frac{6000}{2} = 3000.$$

The number 4000 has digit sum $4 + 0 + 0 + 0 = 4$ (even), so it is not in the set of odd-sum numbers. Therefore, the number of integers strictly greater than 4000 with an odd digit sum is 3000.

Example 39. How many 5-digit numbers divisible by 6 can be formed by the digits 0, 2, 4, 5, 6, 8; repetition of digits not allowed?

[1.0 mark]

Solution 36

The sum of the six digits is $0 + 2 + 4 + 5 + 6 + 8 = 25$. For a 5-digit number to be divisible by 3, the sum of its digits must be a multiple of 3. The excluded digit d must satisfy $25 - d \equiv 0 \pmod 3 \implies 1 - d \equiv 0 \pmod 3 \implies d \in \{4\}$. Thus, the 5 digits must be $\{0, 2, 5, 6, 8\}$. For the number to be divisible by 2, the units digit must be even $(0, 2, 6, 8)$:

- *Units digit is 0:* The first 4 places can be filled by $\{2, 5, 6, 8\}$ in $4! = 24$ ways.
- *Units digit is 2, 6, or 8 (3 choices):* The first place cannot be 0 and cannot be the chosen units digit (3 choices). The remaining 3 middle places can be filled in $3! = 6$ ways. Number of ways $= 3 \times 3 \times 6 = 54$.

Total numbers divisible by $6 = 24 + 54 = 78$.

Question 37

Example 40. Each of the six squares in the strip shown in the figure given below is to be coloured with any one of ten different colours so that no two adjacent squares have the same colour. Find the number of ways of colouring the strip.

[1.0 mark]

Solution 37

For a row of 6 squares $S_1, S_2, S_3, S_4, S_5, S_6$:

- Square 1 can be coloured in 10 ways.
- Square 2 can be any colour except $S_1 \implies 9$ ways.
- Square 3 can be any colour except $S_2 \implies 9$ ways.
- Similarly, squares 4, 5, and 6 each have 9 choices.

$$\text{Total ways} = 10 \times 9^5 = 10 \times 59049 = 590490.$$

Question 38

Example 41. If the letters of the word SACHIN are arranged in all possible ways as listed in a dictionary, then what is the rank of the word SACHIN?

[1.0 mark]

Solution 38

The letters of SACHIN in alphabetical order are:

$$\text{A, C, H, I, N, S.}$$

- Words starting with A: $5! = 120$
- Words starting with C: $5! = 120$
- Words starting with H: $5! = 120$
- Words starting with I: $5! = 120$
- Words starting with N: $5! = 120$

Total words preceding words starting with S $= 5 \times 120 = 600$. The very next word in alphabetical order begins with S followed by the remaining letters in alphabetical order:

$$\text{S A C H I N}$$

Hence, the rank of the word SACHIN is $600 + 1 = 601$.

Question 39

Example 42. Find the number of permutations of n different things taken r at a time such that two specific things occur together. (Exemplar)

[1.0 mark]

Solution 39

Let the two specific items be A and B.

- Since A and B must occur in the arrangement of length r, we need to choose the remaining $r-2$ items from the remaining $n-2$ available items in ${}^{n-2}C_{r-2}$ ways.
- Tie A and B together as a single block (AB). This block together with the other $r-2$ items forms $(r-1)$ entities, which can be arranged in $(r-1)!$ ways.
- Within the block, A and B can be arranged in $2! = 2$ ways.

Thus, the total number of permutations is:

$$ {}^{n-2}C_{r-2} \times (r-1)! \times 2! = 2(r-1) \cdot \frac{(n-2)!}{(n-r)!} = 2(r-1) \cdot {}^{n-2}P_{r-2}. $$